
Question #: 1

A 20-year-old man comes to the physician because of a 2-month history of lack of sweating and flushing. Neurological examination shows damage to postganglionic sympathetic cell bodies. Which of the following structures is most likely damaged?

- A. Brainstem
- B. Sympathetic chain
- C. Dorsal root ganglia
- D. Lateral gray horn
- E. Ventral horn

Question #: 2

A 20-year-old man comes to the emergency department because of a 2-hour history of a painful laceration to the skin of his upper limb. Which of the following fibers most likely transmits the pain sensation to the central nervous system?

- A. Visceral afferent
- B. Somatic efferent
- C. Somatic afferent
- D. Visceral efferent
- E. Autonomic

Question #: 3

A 5-year-old girl is brought to the physician by her mother because of a 1-year history of failure to thrive. Laboratory studies show serum folate levels and blood ammonia levels within the reference ranges. A complete blood picture shows megaloblastic anemia. Urinalysis shows elevated orotic acid levels. Targeted gene mutation analysis shows an inherited deficiency of orotate phosphoribosyltransferase. Which of the following pathways is most likely impaired in this patient?

- A. De novo synthesis of pyrimidine nucleotides
- B. De novo synthesis of purine nucleotides
- C. Formation of uric acid
- D. Catabolism of pyrimidines

E. Salvage of purine nucleotides

Question #: 4

A 40-year-old man comes to the physician because of a 2-week history of itching especially after a warm shower. After a thorough clinical assessment, he is diagnosed with a hematological malignancy called polycythemia vera. He is prescribed hydroxyurea. Which of the following processes is most likely inhibited by this drug?

- A. AMP synthesis from IMP
 - B. Deoxyribonucleotide synthesis
 - C. Uric acid formation from purine nucleotides
 - D. Thymidylate synthesis from dUMP
 - E. Reduction of dihydrofolate to tetrahydrofolate
 - F. GMP formation from IMP
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Question #: 5

A 40-year-old man comes to the physician because of a 1-day history of severe pain and redness in the right great toe. He is prescribed analgesics and a xanthine oxidase inhibitor for long-term management of arthritis. Which of the following pharmacotherapy is most likely prescribed to the patient?

- A. Mycophenolic acid
 - B. Allopurinol
 - C. 5-Fluorouracil
 - D. Quinolones
 - E. Methotrexate
 - F. Sulfonamide
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Question #: 6

A 1-month-old boy is brought to the physician by his parents for a follow-up evaluation. He has had repeated respiratory and gastrointestinal infections since birth. Laboratory studies show a severe reduction in T- and B-cell function. Cytogenomic microarray analysis shows no abnormalities. Neutrophil respiratory burst activity is present on phagocytosis. Which of the following is the most likely diagnosis?

- A. Adenosine deaminase deficiency
- B. DiGeorge syndrome
- C. Orotic aciduria due to UMP synthase deficiency
- D. Chronic granulomatous disease
- E. Lesch Nyhan syndrome

Question #: 7

A 5-year-old boy is brought to the physician by his parents because of a 2-month history of biting his fingers and lips. There is a delay in the developmental milestones. Laboratory studies show serum uric acid is 9.5mg/dL (3-8.2mg/dL). Laboratory studies show very low activity of an enzyme in intermediary metabolism. Which of the following enzyme-catalyzed reactions is most likely impaired in this patient?

- A. Conversion of xanthine to uric acid
- B. Formation of PRPP from ribose 5-phosphate
- C. Formation of deoxyribonucleotides from ribonucleotides
- D. Conversion of adenosine to inosine
- E. Conversion of hypoxanthine to inosine monophosphate (IMP)

Question #: 8

A 25-year-old woman comes to the clinic because of a 1-week history of abdominal pain and loose stools. Physical examination shows tenderness in the left lower quadrant of the abdomen. A complete blood count and liver function tests are within normal limits. Endoscopic examination confirms a diagnosis of Crohn's disease. Treatment is initiated. Two weeks later, her absolute neutrophil count (ANC) decreases from 5140/ μ L to 1010/ μ L, and her white blood cell count (WBC) decreases from 6270/ μ L to 2810/ μ L, prompting a reduction in the dosage of her medication. Genetic analysis shows the patient is homozygous for a mutation in the TPMT gene. Which of the following therapeutic agents is the patient most likely on?

- A. Methotrexate
- B. Azathioprine
- C. 5-fluorouracil

- D. Mycophenolic acid
- E. Hydroxyurea
- F. Trimethoprim

Question #: 9

A 19-month-old boy is brought to the physician by his parents for a follow-up examination. He has a 2-month history of anemia and has been prescribed oral vitamin B12, iron, vitamin B6, and folic acid supplements with no improvement in hematological parameters. Physical examination shows pallor, poorly formed fingernails, and sparse hair. A blood sample is drawn for routine diagnostics. Laboratory studies show megaloblastic anemia. Urinalysis shows elevated orotic acid. Administration of which of the following metabolites will most likely improve this patient's anemia?

- A. Uridine
- B. Deoxyadenosine triphosphate (dATP)
- C. Inosine
- D. Glutamine
- E. Tetrahydrofolate

Question #: 10

A 32-year-old man comes to the physician because of a 4-week history of recurrent episodes of nervousness and palpitations. He says he develops these episodes when he is under stress and each episode lasts for about 15-20 minutes. His pulse is 110/min and blood pressure is 136/84 mm Hg. The physical examination shows anxiety, sweating, fine tremors of hands and dilated pupils. He is diagnosed to have stimulation of the sympathetic nervous system under stress. Which of the following best explains the effects in the target tissues when this division of the nervous system is active in the patient?

- A. The chemical structure of the neurotransmitter
 - B. The properties of the postsynaptic receptor
 - C. The rate of uptake of neurotransmitter
 - D. Amount of calcium released from presynaptic axons
 - E. The velocity of axoplasmic transport in the presynaptic neuron
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Question #: 11

A 34-year-old man comes to the physician because of a 1-week history of difficulty breathing through the nose while sleeping. He says he feels a kind of block in his nose when he goes to supine position. His vital signs are within normal limits. Physical examination shows redness and congestion in the nasal mucosa. He is diagnosed with nasal allergy. He is prescribed a nasal drop that contains an adrenergic receptor agonist, causing vasoconstriction in the nasal blood vessels. Which of the following adrenergic receptor types is the most likely target of this drug?

- A. Alpha-1
- B. Alpha-2
- C. Beta-1
- D. Beta-2
- E. Beta-3

Question #: 12

A 34-year-old man comes to the physician because of a 2-week history of feeling lethargic and drowsy. Physical examination shows constricted pupils, which could result from overactive parasympathetic input. Stimulation of postganglionic parasympathetic neurons is most likely to cause which of the following additional effects?

- A. Tachycardia
- B. Smooth muscle vasodilation
- C. Penile erection
- D. Renin secretion
- E. GI tract wall relaxation

Question #: 13

A 35-year-old woman comes to physician because of a 3-day history of worsening fever and malaise. Her pulse is 105/min, respirations are 18/min and blood pressure is 90/60 mmHg. She is diagnosed with septic shock secondary to an untreated bacterial infection. She is infused with a drug that increased vasoconstriction. Which of the following groups of drugs is most likely administered in this patient?

- A. Alpha 1 adrenergic antagonists
- B. Beta 1 adrenergic antagonists
- C. Alpha 1 adrenergic agonists
- D. Beta 2 adrenergic agonists
- E. Muscarinic (M2) agonists

Question #: 14

A 35-year-old man is brought to the Emergency Department after being severely injured in a car crash. His blood pressure is 50/30 mm Hg and respirations are 25/min. While attempting intubation for surgery, the doctor notices considerable airway secretions. What class of drug would reduce the amount of secretions in this patient?

- A. Alpha 1-adrenoceptor agonist
- B. Nonselective α -adrenoceptor antagonist
- C. Beta 2-adrenoceptor antagonist
- D. Muscarinic receptor agonist
- E. Muscarinic receptor antagonist

Question #: 15

A 50-year-old woman comes to the emergency department because of a 3-hour history of severe chest pain radiating to the left arm. Her pulse is 62/min, blood pressure is 140/86 mm Hg and respirations are 14/min. Physical examination shows profuse sweating, shortness of breath and dizziness. Electrocardiogram and cardiac markers are consistent with acute myocardial infarction. Which of the following best describes the sequence of events activating the sweat glands in this patient?

- A. Postganglionic sympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
- B. Postganglionic parasympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
- C. Preganglionic sympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
- D. Preganglionic parasympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating

- E. Postganglionic sympathetic fibers release norepinephrine, which binds to M3 receptors on the sweat glands to stimulate sweating
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Question #: 16

An investigator is studying the effects of new compounds on arteriolar resistance. Drug X maximally increases vascular resistance by 50% at a dose of 20 mg/mL, and Drug Y maximally increases vascular resistance by 75% at 40 mg/mL. Which of the following best describes the results of the experiment?

- A. Drug X has a smaller volume of distribution than Drug Y
 - B. Drug X has a shorter half-life than Drug Y
 - C. Drug X is less efficacious than Drug Y
 - D. Drug X is less potent than Drug Y
 - E. Drug X has a lower LD50 than Drug Y
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Question #: 17

A 15-year-old boy is brought to the emergency department by his parents because of agitation, mydriasis, hot, dry skin, and tachycardia. Assuming that he is suffering from a drug overdose, which of the following classes of drugs is most likely responsible for his symptoms?

- A. Muscarinic receptor antagonist
 - B. Muscarinic receptor agonist
 - C. Acetylcholinesterase inhibitor
 - D. Beta adrenergic receptor antagonist
 - E. Beta adrenergic receptor agonist
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Question #: 18

An investigator has developed a new drug. The new drug appears to be a partial agonist at receptor X. When the drug is applied to cells expressing receptor X, an influx of sodium is evoked. Receptor X is most likely which of the following?

- A. Alpha1 adrenoreceptor
- B. Beta1 adrenoreceptor

- C. Beta2 adrenoreceptor
- D. Muscarinic cholinergic receptor
- E. Nicotinic cholinergic receptor

Question #: 19

A 25-year-old woman comes to the physician because of a 2-week history of altered vision. She had several episodes where she is beginning to "bump into things around the house". A CT-scan of her brain show areas that are devoid of myelin. Which of the following cells most likely normally produces myelin in this tissue?

- A. Microglia
- B. Astrocytes
- C. Oligodendrocytes
- D. Macrophages
- E. Schwann cells

Question #: 20

A 52-year-old woman comes to the physician because of a 2-week history of pain and muscle weakness to both lower limbs. She had a severe gastrointestinal illness 4 weeks previously. Microscopic studies of the peripheral nerve fibers show an accumulation of lymphocytes, macrophages and plasma cells around nerve fibers and segments of the nerves are found to be devoid of myelin. Which of the following cells normally produce myelin in these nerves?

- A. Oligodendrocytes
- B. Microglia
- C. Schwann cells
- D. Astrocytes
- E. Macrophages

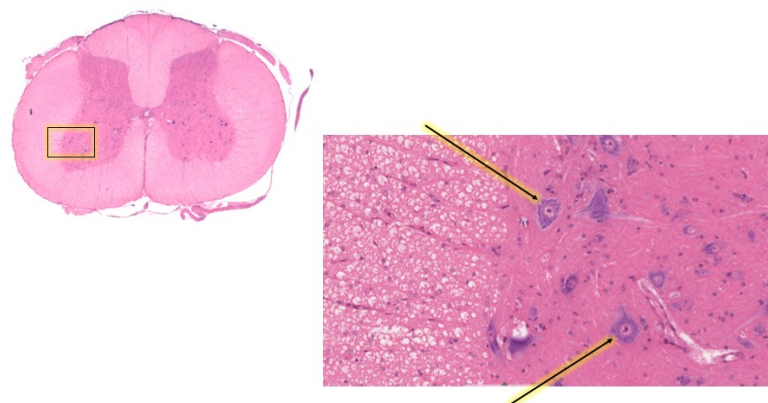
Question #: 21

A 45-year-old man is brought to the emergency department because of back pain following a fall from his roof. Physical examination shows a loss of movement to

the lower limbs. An x-ray of the spine shows a crush injury and fractures to the lumbar level of the vertebral column. An MRI shows the spinal cord to be intact but significantly swollen, characteristic of spinal shock. An image of the spinal cord is attached. Which of the following cells are characteristic of the highlighted region?

- A. Sensory interneurons
- B. Purkinje cells
- C. Pyramidal cells
- D. Alpha motor neurons
- E. Ependymal cells

Attachment:



VentralHorn.jpg

Question #: 22

A research group is investigating the side effects of a new antibiotic on the brain. The drug is administered to rodents over a period of 2 weeks. Laboratory studies show that the drug or its metabolites are present in blood but not in the cerebrospinal fluid. Biopsy studies show that the drug was not present in the brain tissue. Which of the following cells is most likely a component of the structure which prevented the drug from entering the brain tissue?

- A. Schwann
 - B. Ependymal
 - C. Oligodendrocyte
 - D. Astrocytes
 - E. Microglia
-

Question #: 23

A 32-year-old woman comes to the physician because of a 6-month history of lower abdominal pain during her menstrual periods. She has heavy menstrual bleeding during every cycle and intense cramping pain. Physical examination shows no abnormalities. She is diagnosed with menstrual cramps due to the contraction of the smooth muscle of the uterus. Which of the following structures are associated with Ca^{+} delivery to the cytoplasm of these muscle cells?

- A. Condensation bodies
- B. Dense bodies
- C. T-tubules
- D. Caveolae
- E. Intercalated discs

Question #: 24

A 37-year-old man is brought to the emergency department because of a 4-hour history of sudden, severe chest pain. He dies despite appropriate medical care. Autopsy shows a severe form of myocarditis (inflammation of the cardiac muscle). Which of the following best characterizes this type of muscle tissue?

- A. Forms sheets in the wall of hollow organs
- B. Composed of branched, mono or bi-nucleated cells
- C. Composed of long, cylindrical, multi-nucleated cells
- D. Composed of fusiform, mono-nucleated cells
- E. Forms the muscles of the body wall

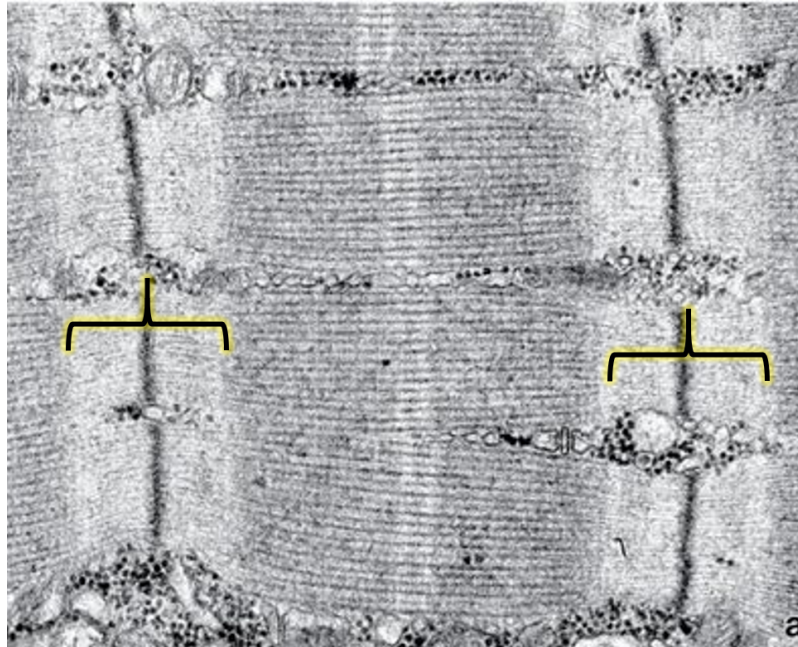
Question #: 25

A 35-year-old man is brought to the emergency department because of recurrent seizures. Physical examination shows tonic-clonic seizure with spasm of the skeletal muscles. Which of the following best characterizes the structure indicated by the brackets on the attached electron micrograph?

- A. Presence of thick filaments only

- B. Presence of thin filaments only
- C. Presence of thick and thin filaments
- D. Absence of any kind of filaments
- E. Makes up a sarcomere

Attachment:



I band Muscle.png

Question #: 26

A 45-year-old man is brought to the emergency department because of sudden, severe chest pain. He dies despite appropriate medical care. Autopsy shows a massive myocardial infarction. Which of the following best characterizes the location of the diad in the type of muscle tissue most likely affected?

- A. At the A-I junction
- B. At the Z-line
- C. In the A-band
- D. Center of H-band
- E. Within the M-line

Question #: 27

A 20-year-old man comes to the physician because of a 1-week history of a painful skin rash. He received a bee sting 2 weeks ago. Physical examination of the

affected area shows redness and mild swelling. A biopsy is taken from the rash and the slides are studied using different stains. One of them was toluidine blue, which brings out the property of metachromasia within certain cells of the connective tissue. Which of the following cells are most likely responsible for this reaction?

- A. Mast cells
- B. Fibroblasts
- C. Macrophages
- D. Neutrophils
- E. Lymphocytes

Question #: 28

A 35-year-old man is admitted to the hospital because of a 10-day history of high fever, chest pain and cough. Physical examination shows a febrile patient with decreased breath sounds on both sides of his chest. An x-ray shows bilateral pneumonia of his lungs and a heart that is located on the right side. A diagnosis of Kartagener's syndrome is made. Which of the following motor proteins are most likely affected in this condition leading to these symptoms?

- A. Actin
- B. Spectrin
- C. Dynein
- D. Myosin
- E. Kinesin

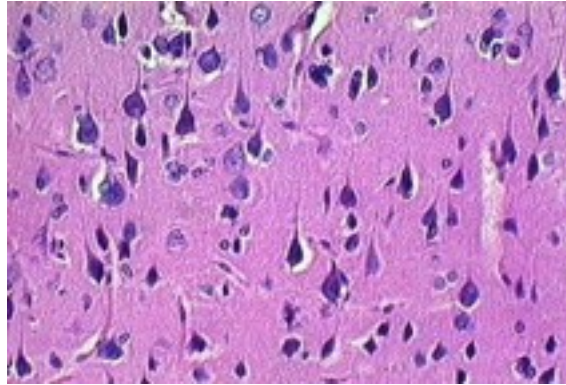
Question #: 29

A 72-year-old man is brought to the physician because of a 3-month history of short and long term memory loss. He is admitted for investigations but soon dies in the hospital. An autopsy is performed to identify the cause of his memory loss and the findings are shown in the attached photomicrograph. Which of the following locations is this biopsy most likely taken from?

- A. Cerebral cortex

- B. Cerebellum
- C. Dorsal root ganglion
- D. Spinal cord
- E. Sympathetic ganglion

Attachment:



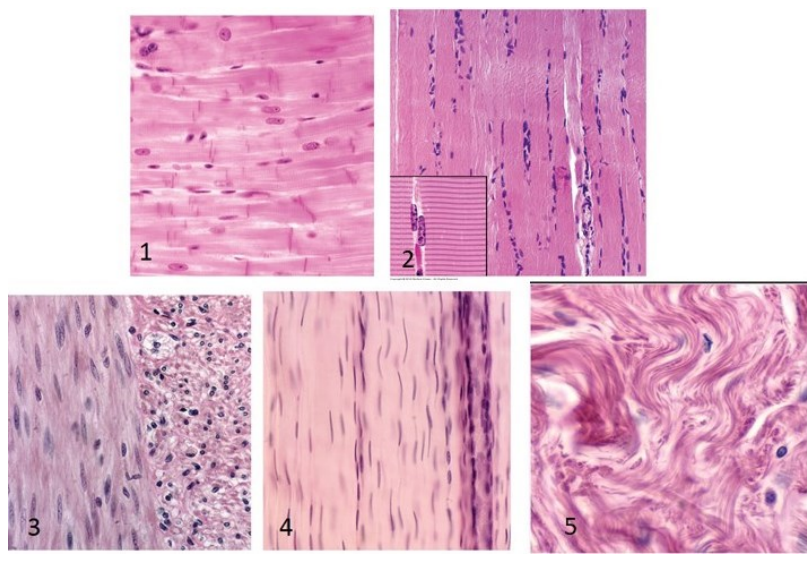
Cerebrum.jpg

Question #: 30

A 34-year-old woman comes to the physician because of a 5-day history of a swelling on her right thigh. Physical examination shows a mass within the extensor muscle compartment of the thigh. A biopsy is obtained and shown in the attached micrograph. Which of the following images best represents the biopsied muscle type?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Attachment:



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